Marco **Valle**

Back-End Developer & Data Analyst

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Summary

Computer Science graduate (BSc, 99/110) specialised in back-end development (Python, Django, SQL) and data analysis (PyTorch, NumPy, pandas). I designed a Django ERP now in production at Aquaforma S.r.l. and **AeroTraj-Ultra**, a non-autoregressive aircraft trajectory prediction model (5–300 s) over 65M real ADS-B records, with an open benchmark over ten models.

Skills

Programming	Python, SQL, C/C++, Kotlin, JavaScript (basic)
Back-End & Web	Django (ORM, migrations), PostgreSQL, SQLite, HTML, CSS
Data Analysis & ML	PyTorch, NumPy, pandas, Matplotlib; OpenSky ADS-B, WGS-84, Gray Code; data pipelines and evaluation
DevOps & Cloud	Git, GitHub Actions (CI), Docker & Docker Compose, Gunicorn, Caddy, Linux VPS deployment
Embedded & Mobile	ESP32/Arduino (C/C++), I2C/ADC sensors, Bluetooth; Android (Kotlin), Gradle
Languages	Italian (native), English (B2, certified)

Education

University of Milano-Bicocca — *Milan, Italy*

BACHELOR'S DEGREE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY (L-31) — *Sep. 2021 – Mar. 2025*

- Graduated with **99/110**. Thesis: *Non-Autoregressive Models for Air Traffic Trajectory Prediction* (PyTorch). **Erasmus+** semester (Oct. 2023 – Mar. 2024) at the University of Łódź.

Experience

Aquaforma S.r.l. — *Treviolo, Italy*

PARTNER AND EMPLOYEE — *2026 – Present*

- Administer and customize **Odoo Enterprise (ERP/CRM)**; IT infrastructure, network, software systems and corporate website.
- Analyse business needs and develop the in-house **ERP in Python/Django**, now the core management system (see Projects).

Pavis S.r.l. — *Rovato, Italy*

SWIMMING POOL INSTALLATION AND MAINTENANCE — *Mar. 2023 – 2025*

- Seasonal maintenance of private pools with water-chemistry analysis (pH, chlorine, cyanuric acid, salinity) and customer and supplier relations.

Poppies S.r.l. — *Milan, Italy*

IT SUPPORT AND CONSULTING — *Feb. 2023 – Jun. 2023*

- Configured on-premise network services; maintained and upgraded workstations and hardware.

Projects

Personal project and open benchmark

AEROTRAJ-ULTRA — NON-AUTOREGRESSIVE AIRCRAFT TRAJECTORY PREDICTION — *2026*

- **Non-autoregressive** multi-horizon model (5–300 s in a single forward pass, **no error accumulation**) anchored to an analytical kinematic propagator on **WGS-84**, with positions encoded in **Gray Code**. Trained on **90–180 consecutive days of real OpenSky traffic** (65M points, 36M windows); 1.28 M parameters, 19.6 ms per aircraft on GPU.
- **Open benchmark**: ten literature models re-implemented and evaluated on the same data with the same protocol (60,000 windows). Mean 3D error of **991.6 m**, the lowest at every horizon.

In production at Aquaforma S.r.l.

BUSINESS MANAGEMENT SYSTEM (ERP) — *2026 – Present*

- Full-stack ERP in **Python/Django + PostgreSQL**: contacts, catalogue, quotes, orders, inventory, delivery notes, invoices, supplier replenishment, **Italian e-invoicing (SDI)** with FatturaPA XML (FPR12) over PEC.
- Deployed on Linux VPS with **Docker Compose** (Gunicorn, Caddy, HTTPS, nightly backups); **600+ automated tests** in **CI on GitHub Actions**; read-only **MCP** server for AI assistants.

Thesis project

FLIGHTPREV — AIRCRAFT TRAJECTORY PREDICTION — *2025 – 2026*

- First model, born from my thesis: **Transformer in PyTorch** on **65M ADS-B records** predicting aircraft positions 10–120 s ahead; average ADE of **582 m against 833 m** for the best kinematic baseline (30% lower) and 75–90% better than **FlightBERT** at all horizons.

Personal project

IOT HOME AUTOMATION & POOL WATER MONITORING — *Ongoing*

- ESP32 firmware (C/C++) for relays and pool water chemistry (I2C, pH with a 16-bit ADC); **Android (Kotlin)** app over Bluetooth.

Extracurricular Activities

Scout Educator, Federazione Scout Europa (Bergamo) since Sep. 2019: educational activities for teens aged 15 to 19, leadership and teamwork.